lecture 1

* Everything we do and work with is a invisible force field

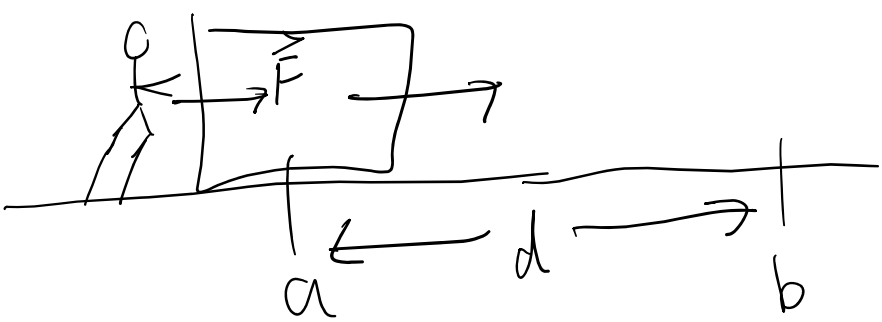
coulomb (C) is the unit of charge

a single electron(q) $= -1.6 \times 10^{-19}\ \frac{C}{\#q}$

$$\frac{1\,C}{\left(1.6\times10^{-19}\frac{C}{q}\right)} = q = \frac{1}{1.6\times10^{-19}} = 6.25\times10^{18} \text{ electrons}$$

Analogy Stand point

• SI units (metric) → French unit

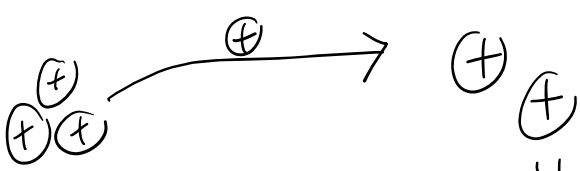


E (energy)

$E = F \cdot d$ [Joules]

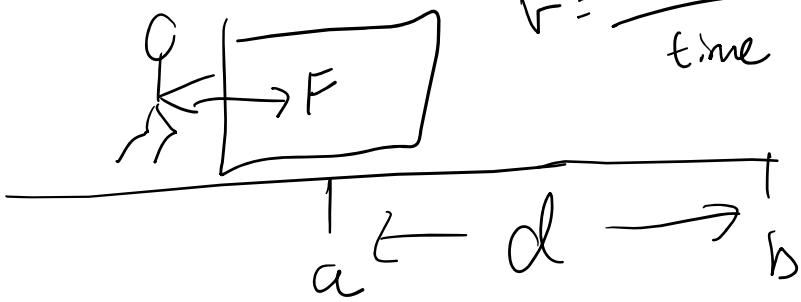
$F = \frac{E}{d} \left[\frac{J}{m}\right]$

EE / CE



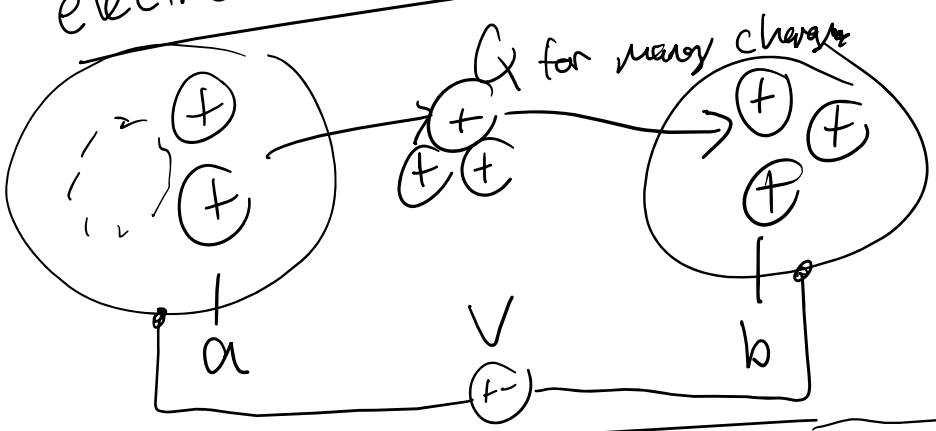
$V = \frac{J}{C}$ [Volts] [V]

its like an electric force



$$v = \frac{\text{distance}}{\text{time}}$$

electric Circuit



$$= \frac{dQ}{dt} = \frac{dQ}{dt} = I \text{ (Amps)}$$

[A]

Dr. V
Conversion where

V = volts

v = velocity

$\mathcal{V}$ = volume

$$\frac{V}{I} = R \text{ (Resistance) } [\Omega]$$

Ohm's law

$$\boxed{V = IR}$$

one of 2 most important eq. in ECE.

$V\left[\frac{J}{C}\right] \cdot I\left[\frac{C}{t}\right] \Rightarrow \left[\frac{J}{t}\right]$ [Watts] Power! [P]

(2) $\boxed{P = V \cdot I}$ 2nd most important equ. in ECE!

(1) $\boxed{V = IR} \rightarrow I = \frac{V}{R}$

$P = (IR)I$ $\qquad P = V\left(\frac{V}{R}\right)$

$\boxed{P = I^2 R}$ $\qquad P = \frac{V^2}{R}$

Anaiden's Computer

$V_{out} = 20V_{out}$

$I_{out} = 12.0\ A$

$240W = P_{out}$

$$\boxed{R_{max} = \frac{V}{I} = \frac{20V}{12A} = 1.7\ \Omega}$$

when the computer is taxed (max output)

Computer is idel

$I = 1A$

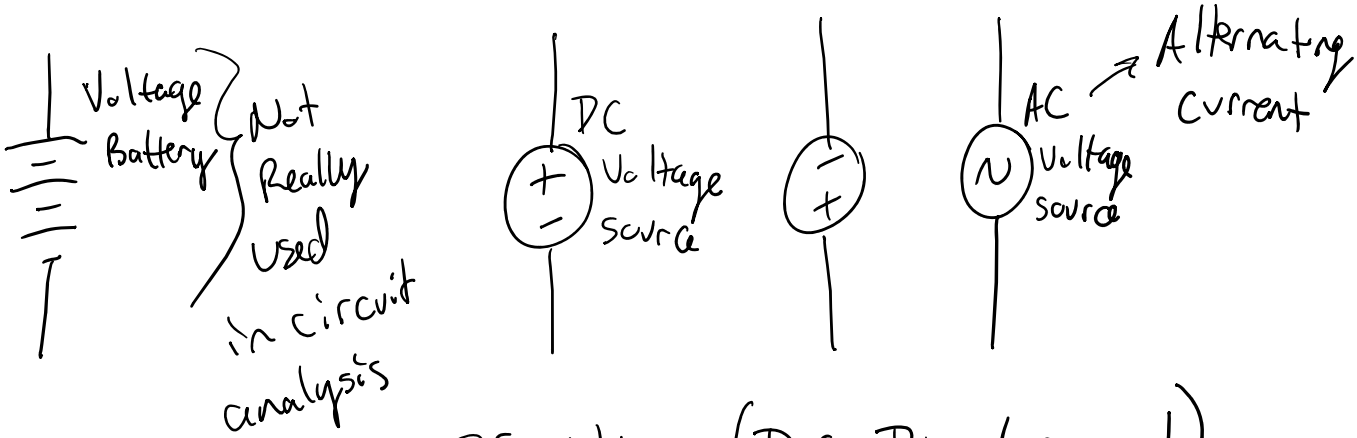
$$\boxed{R_{idel} = \frac{20V}{1A} = 20\Omega}$$

* a computer (what we want to power) is called a Load

A Load (generally) is the Resistor you want to power.

Symbols

Voltage Sources



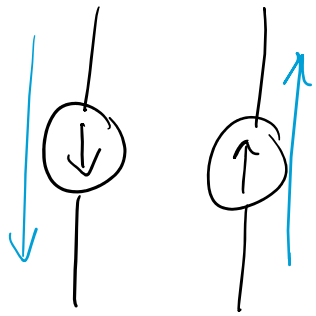
examples DC voltage sources (D.C. Direct current)

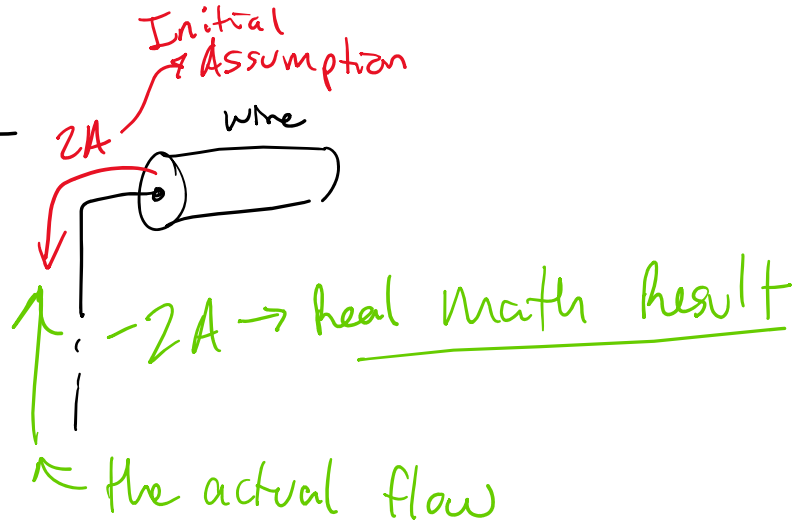
- Batteries
- Phone charger
- Solar Panels

examples of AC source

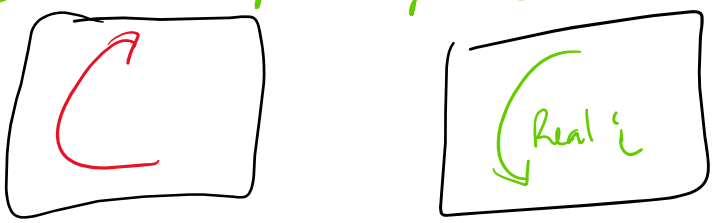
- turbines
- Generators

Current Sources





-i → math gets negative

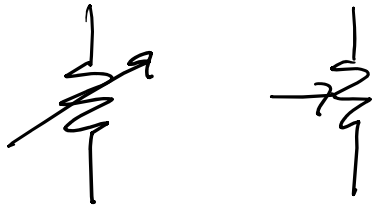


examples of current sources

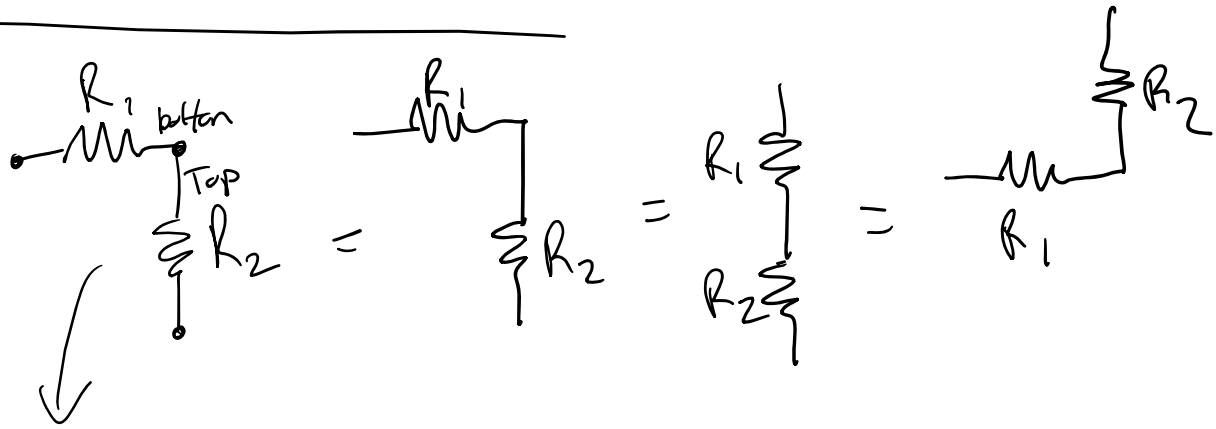
- Solar panels → current source.
- transistors act like a current source.

Resistors

Variable Resistor (Potentiometer) (Potentiostate)

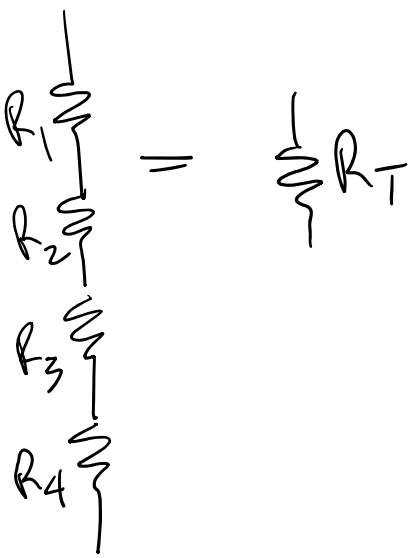


Focus on Resistors

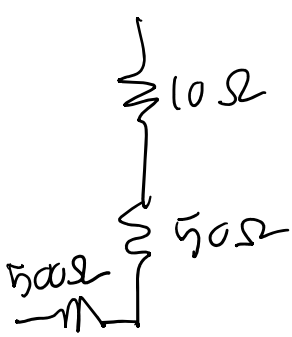


Connection of Resistor bottom to top

Series Connections

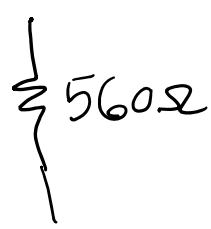


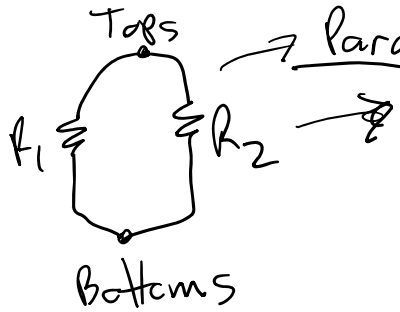
$$R_T = R_1 + R_2 + R_3 + R_4$$



$$R_T = 10 + 50 + 500$$

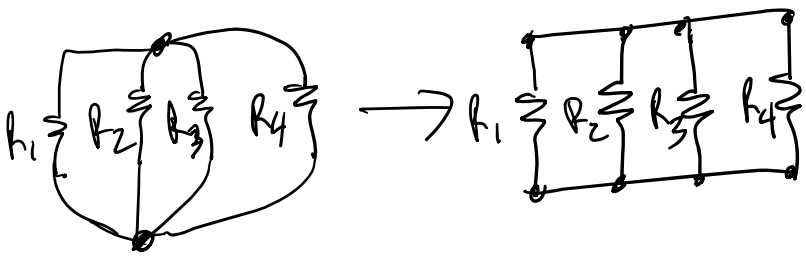
$$\boxed{R_T = 560\Omega}$$





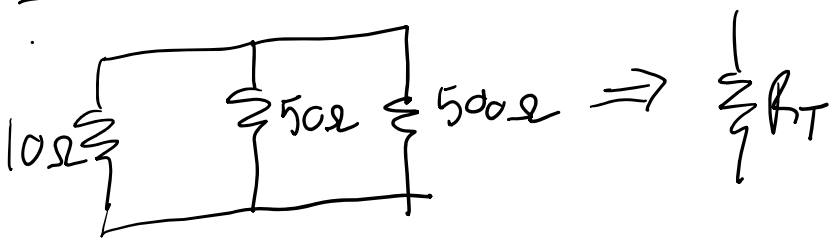
→ Parallel connection

$$R_T = \frac{1}{\left[\left(\frac{1}{R_1}\right) + \left(\frac{1}{R_2}\right) + \left(\frac{1}{R_3}\right) + \cdots\right]}$$



$R_1 = R_2 = R_3 = R$ if all Resistors are equal.

$$R_T = \frac{1}{\left(\frac{1}{R} + \frac{1}{R} + \frac{1}{R}\right)} = \frac{1}{\left(\frac{3}{R}\right)} = \boxed{\frac{R}{3}}$$

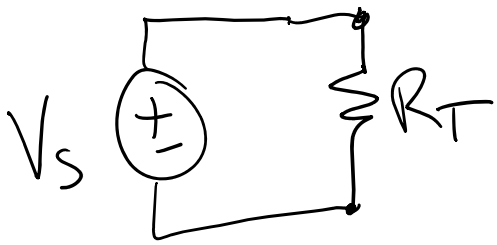


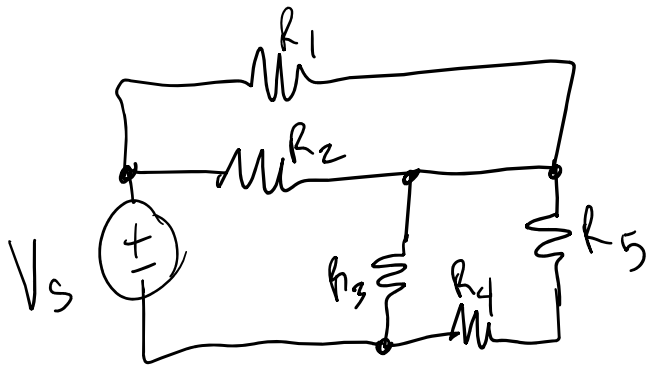
$$R_T = \frac{1}{\left(\frac{1}{10} + \frac{1}{50} + \frac{1}{500}\right)}$$

$$\boxed{R_T = 8.2\,\Omega}$$

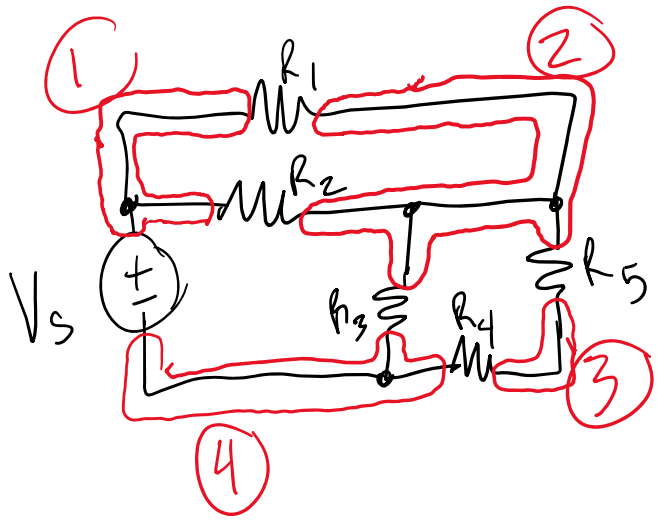
Circuit Reduction means

all circuit can be Reduced to.



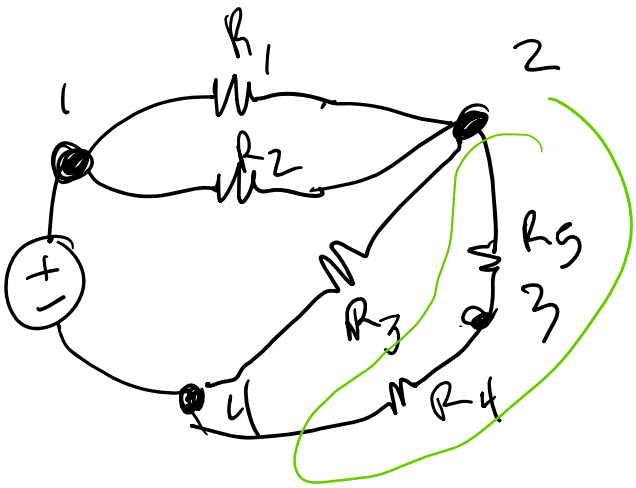


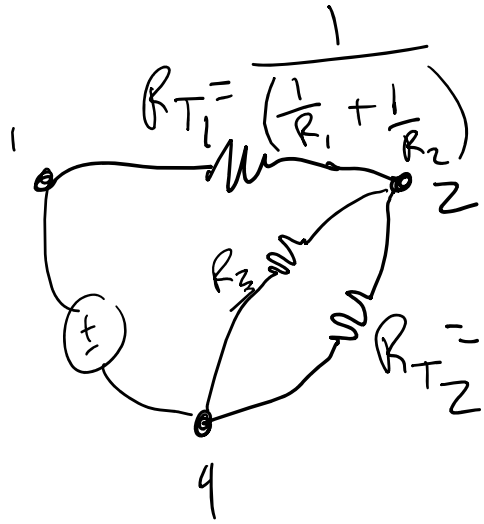
Circuit Strategies for Reduction



1. Identification Nodes

* the bubble trick





$$R_{T_1} = \frac{1}{\left(\frac{1}{R_1} + \frac{1}{R_2}\right)}$$

$$R_{T_2} = R_5 + R_4$$

$$R_3 \| R_{T_2}$$

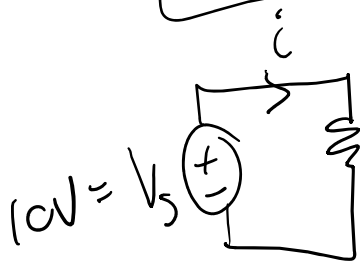
↑

in parallel

$$R_{T_3} = R_3 \| R_{T_2}$$

$$R_{T_4} = R_{T_1} + R_{T_3}$$

$0.05\,\Omega$



$V = IR$

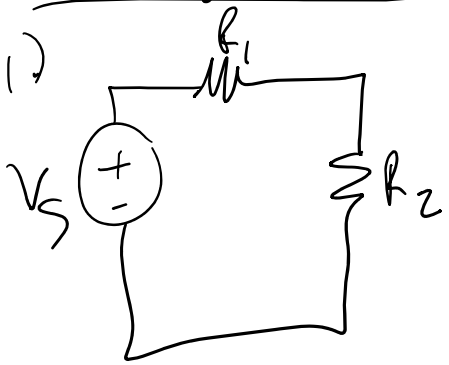
$$\frac{10}{0.5} = I = 20A$$

$20A \cdot 10V = P$

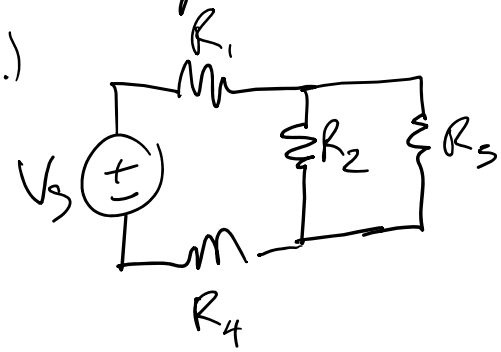
$200W = P$ ✓

Identify the number of Nodes (Activity Class)

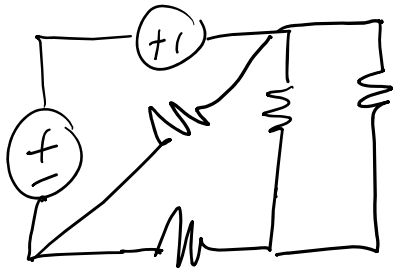
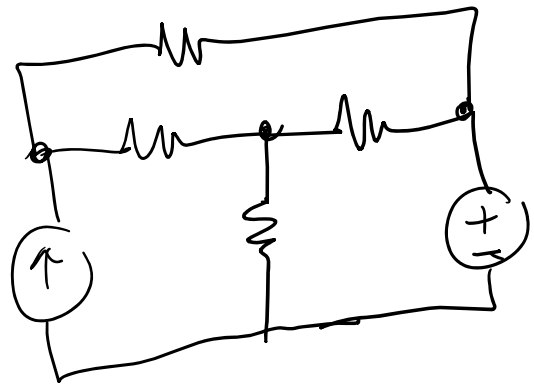
1.)



2.)



3.)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